

PURGUARD: Purification-Aware Voice Protection Against Diffusion-Based Removal Attacks

Anonymous Authors

Abstract—Protective perturbations are a promising proactive defense against unauthorized voice cloning, yet attacks using diffusion purifiers can remove these perturbations before cloning, bypassing existing defenses. This gap arises because most protection methods optimize against the cloning model directly, treating purification only as an evaluation condition. We propose PURGUARD, a purification-aware voice protection framework that incorporates the attacker’s diffusion purifier into the optimization loop. PURGUARD jointly maximizes post-purification speaker-identity disruption and trajectory deviation inside the reverse diffusion process, making protected waveforms resistant to purification-based recovery. We operationalize the framework with purifier-in-the-loop optimization and ensemble-based speaker-identity objectives, and define a benchmark protocol covering direct protection, post-purification robustness, clean-purification controls, and perceptual quality. Our formulation reframes robust voice anti-cloning as an adaptive problem against the attacker’s purification path, rather than optimizing directly against the cloning backend alone.

Index Terms—voice cloning, protective perturbations, diffusion purification, adversarial robustness, speaker identity

1. Introduction

Modern speech synthesis systems can clone a speaker from only a small amount of reference audio. This capability enables useful applications such as dubbing, accessibility, and personalization, but it also creates a practical security risk. In 2019, fraudsters reportedly used AI-generated speech to impersonate a corporate executive and steal more than \$243,000 over a phone call [1]. In 2021, another reported voice-cloning heist led to a \$35 million bank transfer [2]. Attackers have also demonstrated that AI-generated voices can break into voice-protected bank accounts [3]. These incidents reflect the same operational path studied by recent voice-protection work: speech uploaded to public platforms can be collected, cloned through open-source or commercial synthesis services, and reused for impersonation, social engineering, telecom fraud, and voice-biometric abuse [4]–[6]. Because the defender often controls the audio only before publication, proactive protection at the data source is an important line of defense.

Protective perturbation methods address this setting by adding small, human-tolerable perturbations to the released

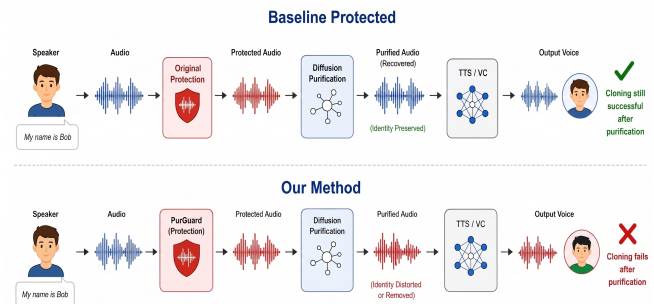


Figure 1. High-level attack chain for purification-aware voice protection: the defender releases protected speech, while the attacker applies a diffusion purifier before voice cloning.

waveform so that downstream voice cloning or speaker modeling systems extract unreliable identity information [4]–[8]. This line of work has advanced quickly: early methods attacked voice conversion or zero-shot cloning, while newer systems consider broader synthesis scenarios, stronger transferability, and perceptual constraints. Under a direct-cloning protocol, these defenses can make synthesized speech less similar to the source speaker.

The direct-cloning protocol, however, is no longer sufficient. We call the stronger threat a *purification-removal attack*: after obtaining a protected public recording, the attacker first applies a diffusion purifier to remove protective perturbations and then feeds the recovered reference to a cloning backend. De-AntiFake and VocalBridge show that this purification stage can be inserted before cloning, using diffusion or bridge-diffusion models to remove protective perturbations and recover speaker-discriminative cues [9], [10]. In this attack chain, the attacker does not ask whether the protected waveform is immediately cloneable; the attacker first asks whether purification can convert it back into a useful reference. A defense that is optimized only for the unpurified input may therefore overstate user protection.

This paper studies the defender’s side of the same arms race. The central gap is that existing voice protection methods usually treat purification as an external evaluation condition rather than as a differentiable adversary in the optimization loop. Meanwhile, DiffAttack shows that output-only supervision can be weak and proposes intermediate-step supervision over the diffusion trajectory, while Diff-Break shows that stochastic and adaptive evaluation can

overturn apparent robustness claims for diffusion purification [11], [12]. Voice anti-cloning is different: the target is a continuous speaker identity representation, the perturbation must remain suitable for public release, and the downstream misuse pipeline may contain unknown cloning backends.

We propose PURGUARD, a purification-aware framework for protective perturbation generation. Figure 1 summarizes the pipeline. The key idea is simple: if the attacker depends on a purifier to recover speaker identity, then the defender should optimize the perturbation against the purifier, not only against the final cloning model. PURGUARD combines a speaker-disruption objective, which reduces similarity between the source speaker and the purified protected speech, with a trajectory-deviation objective, which pushes the reverse diffusion path away from its paired forward noising states. This makes purification resistance a training objective rather than an after-the-fact stress test.

Our implementation instantiates this idea with the first-stage DiffWave purifier from De-AntiFake, a differentiable ensemble of VITS, Tortoise, WavLM, and ECAPA-style speaker encoders when available, and independent ECAPA, x-vector, and Resemblyzer evaluators. We also define a purification-aware evaluation protocol with clean-purification controls, post-purification speaker similarity, thresholded speaker-verification acceptance, and quality metrics. The protocol is designed to answer the security question that matters in practice: whether protected speech remains non-cloneable after the attacker attempts to purify it.

Our contributions are:

- We formulate *purification-aware voice protection*, in which the defender must preserve speaker disruption even after an attacker applies diffusion-based perturbation removal.
- We introduce PURGUARD, a dual-objective optimization framework that jointly maximizes post-purification speaker-identity disruption and trajectory deviation inside the attacker’s reverse diffusion process.
- We define a purification-aware evaluation protocol with clean-purification controls, calibrated speaker-verification thresholds, and repeated stochastic purification trials, enabling fair comparison across protection methods and diffusion purifiers.

2. Related Work

2.1. Proactive Voice Protection

Proactive voice protection aims to modify speech before release so that unauthorized synthesis systems cannot reuse the speaker identity. AttackVC perturbs source speech to disrupt voice conversion [7], VoiceGuard improves imperceptibility with time-domain constraints [8], and AntiFake improves transferability by attacking speaker representations used by cloning systems [4]. SafeSpeech expands the scenario to training-time malicious synthesis and emphasizes

robust, universal perturbations [5]. E2E-VGuard further considers end-to-end and LLM-based synthesis pipelines, where automatic transcription and timbre extraction jointly affect cloning quality [6]. These systems motivate our goal but share a common limitation: they optimize against the cloning or synthesis pipeline directly, and do not account for an attacker who first removes the perturbation with a diffusion purifier. PURGUARD addresses this gap by treating purification as a first-class component of the optimization target.

2.2. Purification as an Adaptive Attacker

Adversarial purification removes perturbations from the input before the downstream model consumes it. AudioPure shows that DiffWave-style forward noising followed by reverse denoising can defend audio recognition models against adversarial examples [13]. Later systems improve purification in different ways, including hierarchical diffusion [14], efficient dual-domain purification [15], and speaker-verification-oriented diffusion reconstruction [16]. These methods are valuable to an attacker in the voice anti-cloning setting because they can be used as a preprocessing step before synthesis.

De-AntiFake directly evaluates this risk for voice cloning and shows that purification can bypass several protective perturbation defenses [9]. VocalBridge strengthens this line by moving purification into a latent diffusion-bridge formulation that aims to recover clean, speaker-discriminative speech while avoiding transcript dependence [10]. These works expose the fragility of current protection methods under adaptive attackers. PURGUARD takes the complementary defender perspective: rather than building a stronger purifier, it generates perturbations that are optimized to resist purification, closing the loop between attack and defense.

2.3. Adaptive Attacks on Diffusion Purification

The conceptual foundation of PURGUARD is related to adaptive attacks on diffusion purification. DiffAttack shows that output-only supervision can be weak and proposes intermediate-step supervision over the diffusion trajectory [11]. DiffBreak shows that stochastic and adaptive evaluation can overturn apparent robustness claims for diffusion purification [12]. These studies suggest two lessons for voice protection: the purifier’s internal trajectory is an attack surface, and single-pass output evaluation can misrepresent robustness.

The most closely related work is AntiPure, which studies protective perturbations against diffusion purifiers for image customization [17]. PURGUARD borrows the trajectory-aware view from this line of work, but applies it to a different security objective: protecting released speech from post-purification voice cloning under perceptual audio-quality constraints and continuous speaker-identity metrics.

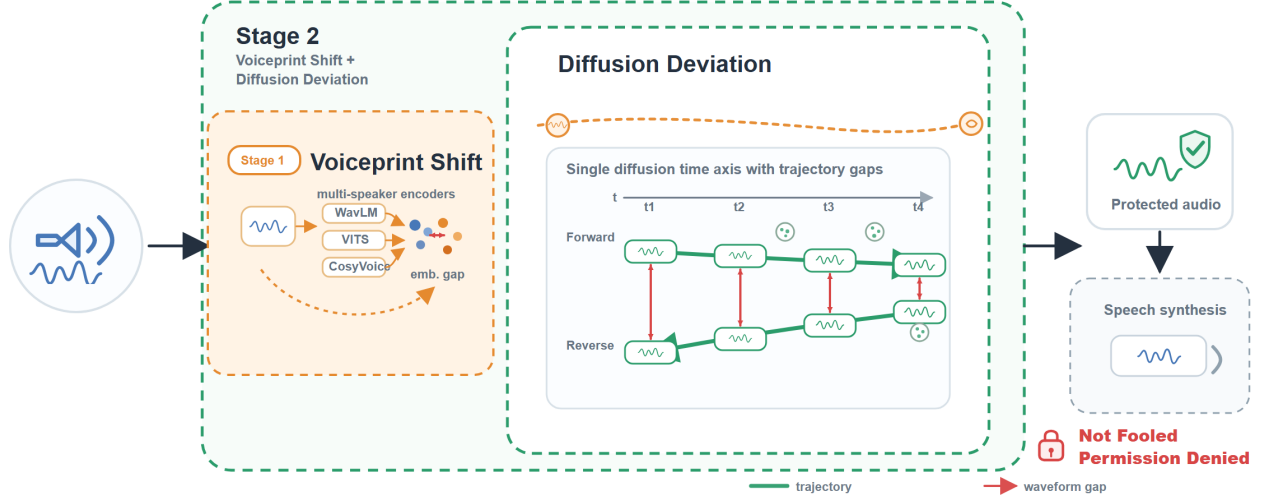


Figure 2. PURGUARD framework. The defender optimizes a bounded protective perturbation through a surrogate diffusion purifier, combining post-purification speaker-identity disruption with reverse-trajectory deviation so that purified protected speech remains unreliable for downstream cloning.

3. The PURGUARD Framework

3.1. Threat Model and Security Goal

Figure 2 gives an overview of the PURGUARD framework. We study a proactive defense for public speech release. Let $x \in [-1, 1]^T$ be a clean waveform from a protected speaker. The defender releases

$$x_p = \text{clip}_{[-1, 1]}(x + \delta), \quad \|\delta\|_\infty \leq \epsilon, \quad (1)$$

where the perturbation budget keeps x_p close to the original audio. The attacker wants to use the released audio as a reference for voice cloning, but first applies a diffusion purifier $\mathcal{P}$ to remove protective perturbations. The effective attack chain is therefore

$$x_p \rightarrow \hat{x} = \mathcal{P}(x_p) \rightarrow \mathcal{C}(\hat{x}), \quad (2)$$

where $\mathcal{C}$ is a downstream cloning or voice-conversion system.

The defender’s goal is stronger than direct anti-cloning. A protected waveform should remain non-cloneable after the attacker purifies it. We call this property *purification-resilient protection*: the purified protected reference $\mathcal{P}(x_p)$ should not preserve enough source-speaker information for successful cloning, while the protected waveform remains intelligible to human listeners.

We study a black-box target-purifier setting. The defender does not know which purifier the attacker will deploy

at test time and cannot query or differentiate through the attacker’s target purifier during perturbation generation. To construct the perturbation, the defender uses a local surrogate purifier and differentiates through its denoising process. The security question is therefore not whether PURGUARD overfits to a known purifier, but whether purification-aware perturbations learned from a surrogate transfer to unseen target purifiers used by the attacker.

In practice, the defender may not know which purifier the attacker will use. We address this in two ways. First, we evaluate transfer to held-out target purifiers (AudioPure, DualPure, WavePurifier) that were not used during optimization, measuring how much protection degrades when the surrogate does not match the attacker’s purifier. Second, the trajectory-deviation objective targets the structural behavior of diffusion reverse processes rather than the specific weights of one model, which we hypothesize improves cross-purifier transfer compared to output-only optimization. The remaining open problem is a no-surrogate setting, where the defender has no local purifier at all during optimization; we discuss this limitation in Section 5.

3.2. Design Rationale

Existing voice protection methods typically optimize the released waveform against a cloning model or a speaker encoder. This direct objective can be effective when the attacker uses x_p immediately. It becomes incomplete when

the attacker first runs a purifier. In that setting, the important object is not only x_p , but the cleaned reference $\mathcal{P}(x_p)$ that the cloning system actually sees.

PURGUARD changes the defense target from the final cloning model to the full purification attack chain. The defense has two requirements. First, the purified audio should lose source-speaker identity according to speaker encoders that approximate cloning-relevant identity extraction. Second, the perturbation should interfere with the purifier’s denoising process rather than behave like ordinary noise that the purifier can remove. These requirements lead to two complementary objectives: post-purification identity disruption and reverse-trajectory deviation.

3.3. Purification-Aware Objective

PURGUARD optimizes a bounded perturbation by solving

$$\delta^* = \arg \max_{\|\delta\|_\infty \leq \epsilon} \alpha \mathcal{L}_{\text{id}}(x_p; \mathcal{P}, \mathcal{F}) + \beta \mathcal{L}_{\text{traj}}(x_p; \mathcal{P}), \quad (3)$$

where $\mathcal{F} = \{f_k\}_{k=1}^K$ is an ensemble of differentiable speaker encoders. The weights α and β control the balance between identity disruption and purification resistance. Eq. (3) is the central methodological choice: purification is modeled as part of the adversary, not as a post-hoc evaluation step.

3.4. Post-Purification Identity Disruption

For each encoder f_k , let $e_k = f_k(x)$ be the source speaker embedding. In the untargeted setting, PURGUARD reduces source-speaker similarity after purification:

$$\mathcal{L}_{\text{id}} = -\frac{1}{K} \sum_{k=1}^K \cos(f_k(\mathcal{P}(x_p)), e_k). \quad (4)$$

This objective directly matches the attacker’s use case. The attacker does not clone from the raw protected waveform; the attacker clones from the recovered reference $\mathcal{P}(x_p)$. Therefore, the identity objective is evaluated after purification.

The encoder ensemble reduces overfitting to one identity extractor. A single encoder can produce perturbations that exploit that encoder while transferring poorly to other cloning systems. Averaging over multiple encoders encourages perturbations that disrupt speaker information shared across common identity representations. Evaluation still uses held-out encoders, so success cannot be explained only by access to the optimization ensemble.

3.5. Reverse-Trajectory Deviation

Optimizing only Eq. (4) can be weak because the gradient must pass through stochastic noising, multiple denoising steps, and a speaker encoder. More importantly, a final-output objective gives the purifier many opportunities to remove the perturbation before the identity loss is measured.

PURGUARD therefore also constrains intermediate denoising behavior.

The rationale follows a key observation from adaptive attacks on diffusion purification. DiffAttack shows that diffusion purifiers are vulnerable when an input can make the reverse trajectory reconstruct states that disagree with the corresponding forward diffusion states; this deviated reconstruction induces inaccurate density-gradient estimates and causes the sampler to leave the intended clean-data reconstruction path [11]. We use the same mechanism from the defender’s side. A successful purifier should convert a protected reference into a clean, speaker-preserving reference by treating the protection signal as removable noise. If the perturbation instead changes the purifier’s reverse trajectory, the purifier no longer behaves as a benign denoiser for speaker identity: its intermediate states become inconsistent with the identity-preserving path, and the final purified waveform can remain intelligible while losing speaker cues needed for cloning.

For a timestep t , define the paired forward noising state

$$\begin{aligned} x_t^+ &= \sqrt{\alpha_t} x_p + \sqrt{1 - \alpha_t} z, \\ z &\sim \mathcal{N}(0, I). \end{aligned} \quad (5)$$

Starting from the terminal noisy state, the purifier produces reverse states x_t^- along its denoising path. For captured timesteps $\mathcal{T}$, PURGUARD maximizes

$$\mathcal{L}_{\text{traj}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|x_t^- - x_t^+\|_2^2. \quad (6)$$

Here, the superscript $+$ denotes the paired forward noising state of the protected waveform, while the superscript $-$ denotes the reverse denoising state produced by the purifier at the same timestep.

Proposition 1 (Trajectory loss lower-bounds clean-path deviation). Let $x_t^{0,+} = \sqrt{\alpha_t} x + \sqrt{1 - \alpha_t} z$ be the forward noising state of the clean waveform under the same Gaussian noise z used in x_t^+ . If $\|x_p - x\|_2 \leq \rho$, then for any reverse state x_t^- produced by the purifier,

$$\|x_t^- - x_t^{0,+}\|_2 \geq \|x_t^- - x_t^+\|_2 - \sqrt{\alpha_t} \rho. \quad (7)$$

Consequently, over captured timesteps $\mathcal{T}$,

$$\begin{aligned} &\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|x_t^- - x_t^{0,+}\|_2^2 \\ &\geq \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \left([\|x_t^- - x_t^+\|_2 - \sqrt{\alpha_t} \rho]_+ \right)^2, \end{aligned} \quad (8)$$

where $[a]_+ = \max(a, 0)$. Thus, once the trajectory loss exceeds the small radius induced by the perturbation budget, Eq. (6) certifies a positive lower bound on how far the purifier’s reverse states deviate from the clean forward noising path that a speaker-preserving reconstruction should track.

Proof. Because $x_p = x + \delta$, the paired protected and clean forward states under the same noise differ by

$$x_t^+ - x_t^{0,+} = \sqrt{\alpha_t} \delta. \quad (9)$$

Therefore $\|x_t^+ - x_t^{0,+}\|_2 \leq \sqrt{\alpha_t}\rho$. Applying the reverse triangle inequality gives

$$\|x_t^- - x_t^{0,+}\|_2 \geq \|x_t^- - x_t^+\|_2 - \|x_t^+ - x_t^{0,+}\|_2, \quad (10)$$

which proves Eq. (7). Squaring the nonnegative part of the right-hand side and averaging over $\mathcal{T}$ proves Eq. (8). DiffAttack gives a complementary density-estimation interpretation of the same mechanism: deviating reverse states from their paired forward states induces inaccurate score or density-gradient estimates in diffusion purification [11]. In PURGUARD, Eq. (6) supplies this trajectory-level pressure, while Eq. (4) aligns the pressure with the speaker-identity property we need to disrupt.

The role of Eq. (6) is not to make the released waveform noisy. Its role is to make the purifier’s intermediate reconstruction states disagree with the protected input’s paired noising path. A diffusion purifier succeeds only if its reverse process repeatedly estimates directions that remove perturbation while preserving the speaker cues needed for cloning. By maximizing trajectory deviation, PURGUARD makes these intermediate estimates less reliable: the perturbation is optimized to interfere with the reverse denoising path itself, so the purifier cannot simply project the protected audio back to a speaker-preserving reference. This gives the perturbation a target inside the purifier, rather than relying only on the final purified waveform.

Because the diffusion reverse process is stochastic, we optimize this trajectory objective under expectation over transformation (EOT), rather than against a fixed noising path. During optimization, the Gaussian noise z and the purifier’s stochastic reverse trajectory are resampled across iterations, so the perturbation is trained against the expected purification behavior instead of overfitting to one random seed.

3.6. Practical Optimization

We solve Eq. (3) with projected gradient ascent using the staged procedure in Algorithm 1. The first stage is a speaker-encoder warm start. It optimizes only the direct embedding displacement

$$\mathcal{L}_{\text{dir}}(x_p; \mathcal{F}) = -\frac{1}{K} \sum_{k=1}^K \cos(f_k(x_p), e_k), \quad (11)$$

which moves the protected waveform away from the source speaker in the ensemble embedding space without paying the diffusion cost. The second stage switches on the full purification-aware objective: it maximizes trajectory MSE between paired forward and reverse diffusion states while also maximizing post-purification embedding displacement from the original speaker.

The switch from the warm-start stage to the purification-aware stage is fixed by the iteration budget, not by a convergence test. At iteration i ,

$$\begin{aligned} \tilde{x}_p^{(i+1)} &= \text{clip}_{[-1,1]} \left(x_p^{(i)} + \eta \text{sign}(\nabla_{x_p} \mathcal{L}) \right), \\ x_p^{(i+1)} &= \Pi_{x,\epsilon} \left(\tilde{x}_p^{(i+1)} \right), \end{aligned} \quad (12)$$

Algorithm 1 PURGUARD staged optimization

- 1: **Input:** clean waveform x , encoder ensemble $\mathcal{F}$, surrogate diffusion purifier $\mathcal{P}$, budget ϵ , total steps N , warm-start steps N_1 , step size η , weights α, β , captured timesteps $\mathcal{T}$
 - 2: Initialize $x_p^{(0)} \leftarrow \text{clip}_{[-1,1]}(x + u)$, where $u \sim \mathcal{U}(-\epsilon, \epsilon)$
 - 3: Precompute source embeddings $e_k = f_k(x)$ for all $f_k \in \mathcal{F}$
 - 4: **for** $i = 0$ to $N - 1$ **do**
 - 5: **if** $i < N_1$ **then**
 - 6: $\mathcal{L} \leftarrow \mathcal{L}_{\text{dir}}(x_p^{(i)}; \mathcal{F})$
 - 7: **else**
 - 8: Sample fresh diffusion noise z and reverse randomness
 - 9: Build paired forward states x_t^+ and reverse states x_t^- for $t \in \mathcal{T}$
 - 10: $\hat{x} \leftarrow \mathcal{P}(x_p^{(i)})$
 - 11: $\mathcal{L}_{\text{id}} \leftarrow -\frac{1}{K} \sum_{k=1}^K \cos(f_k(\hat{x}), e_k)$
 - 12: $\mathcal{L}_{\text{traj}} \leftarrow \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|x_t^- - x_t^+\|_2^2$
 - 13: $\mathcal{L} \leftarrow \alpha \mathcal{L}_{\text{id}} + \beta \mathcal{L}_{\text{traj}}$
 - 14: **end if**
 - 15: $\tilde{x} \leftarrow \text{clip}_{[-1,1]} \left(x_p^{(i)} + \eta \text{sign}(\nabla_{x_p} \mathcal{L}) \right)$
 - 16: $x_p^{(i+1)} \leftarrow \Pi_{x,\epsilon}(\tilde{x})$
 - 17: **end for**
 - 18: **return** $x_p^{(N)}$
-

where $\Pi_{x,\epsilon}$ projects to the ℓ_∞ ball around x .

The implementation differentiates through reverse steps separately and accumulates gradients, which avoids storing the full diffusion graph. We implement EOT by drawing fresh diffusion noise and reverse-process randomness during iterative optimization; the resulting gradient is a Monte Carlo estimate of the purification-aware objective. We use a fixed warm-start budget N_1 followed by purification-aware updates for the remaining iterations, with exact hyperparameters reported in the evaluation configuration. The surrogate purifier is instantiated with De-AntiFake’s first-stage DiffWave model [9].

3.7. Adaptive Adversary Discussion

An adaptive attacker may change purification steps, random seeds, purifier architecture, or the downstream cloning backend. PURGUARD is designed to make this adaptation harder in three ways. First, the identity loss is measured after purification, so the defense targets the attacker’s recovered reference rather than the raw protected audio. Second, the trajectory loss supervises intermediate states, which reduces dependence on a single final-output gradient. Third, the encoder ensemble and held-out evaluation separate the optimization models from the models used to judge cloning risk.

These choices do not make purification-resistant protection absolute. A sufficiently strong attacker could train a purifier against PURGUARD-generated examples, combine

multiple purifiers, or use human-in-the-loop editing. We therefore evaluate PURGUARD as a robustness improvement under a stronger attack chain, not as a complete solution to voice impersonation. This framing matches the security goal of reducing the reliability of unauthorized cloning from public audio under realistic adaptive pressure.

4. Evaluation

4.1. Research Questions

The evaluation is organized around four questions. **RQ1:** Do existing protective perturbations remain effective when an attacker purifies the released audio before cloning? **RQ2:** Is speaker-only optimization sufficient, or does explicit trajectory supervision improve post-purification protection? **RQ3:** Does protection transfer across purifiers and independent speaker evaluators, rather than overfitting to one surrogate encoder? **RQ4:** Does purification resistance come at an unacceptable perceptual-quality cost?

We evaluate on VoxCeleb1 [18] using 800 held-out utterances. Following DiffBreak’s finding that stochastic and adaptive evaluation can overturn apparent robustness claims for diffusion purification, all metrics are averaged over three independent purification seeds [12].

4.2. Compared Systems and Attack Chain

Protection methods.. The benchmark is designed to compare **AttackVC** [7], **VoiceGuard** [8], **AntiFake** [4], **SafeSpeech** [5], **E2E-VGuard** [6], and PURGUARD. We treat all of them as proactive protection methods and evaluate how much source-speaker information remains after the attacker purifies and clones the protected reference. Internal variants of PURGUARD, including speaker-only, trajectory-only, and staged joint optimization, are used for ablation.

Diffusion purifiers.. The surrogate purifier used for optimization is the first-stage DiffWave model from **De-AntiFake** [9]. At evaluation time, we treat the attacker’s target purifier as black-box to the defender and test transfer to **VocalBridge** [10], **WavePurifier** [14], **AudioPure** [13], and **DualPure** [15]. These purifiers differ in representation, diffusion design, and reconstruction assumptions, making them useful black-box transfer tests for purification-aware protection.

Cloning backends and evaluators.. After purification, the attacker uses the recovered reference in representative synthesis backends: CosyVoice, F5-TTS, and Chatterbox. Speaker similarity is measured by independent evaluators, including ECAPA-TDNN [19], x-vector, and Resemblyzer/d-vector style encoders. This design separates the encoder used to optimize PURGUARD from the encoders used to evaluate protection.

Evaluator	EER	Threshold
ECAPA-TDNN	0.0037	0.3195
x-vector	0.0600	0.9472
Resemblyzer	0.0425	0.7235

TABLE 1. SPEAKER-VERIFICATION THRESHOLD CALIBRATION USED BY THE CURRENT EVALUATION IMPLEMENTATION. THRESHOLDS ARE APPLIED PER EVALUATOR WHEN COMPUTING SVA AND PPR.

4.3. Metrics and Controls

The primary continuous metric is source-to-output speaker similarity:

$$\text{Sim}_{g,m} = \cos(g(x), g(\mathcal{C}_m(\mathcal{P}(x_p)))) , \quad (13)$$

where lower values indicate stronger retained protection. We also report speaker verification acceptance (SVA), attack success rate (ASR), post-purification protection rate (PPR), and quality metrics including SNR, PESQ, and STOI. Following De-AntiFake [9], SVA is the average binary speaker-verification acceptance rate over cloned outputs:

$$\text{SVA}_{g,m,p} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\cos(g(x_i), g(\mathcal{C}_m(p(x_{p,i})))) \geq \tau_g] , \quad (14)$$

where g is the verification evaluator, m is the cloning backend, p is the purification setting, and τ_g is the calibrated acceptance threshold. We report **xSVA** when g is an x-vector verifier and **dSVA** when g is a d-vector-style verifier. Higher xSVA/dSVA means the attacker more often produces speech accepted as the source speaker; lower xSVA/dSVA means stronger retained protection. The most important controls are: clean reference, protected reference without purification, purified clean reference, and purified protected reference. The purified-clean control is essential because it verifies that a purifier preserves identity for benign inputs.

4.4. Threshold Calibration

The implementation includes calibrated decision thresholds for the independent speaker evaluators. Table 1 records the current calibration values used by the evaluation code. These thresholds convert continuous cosine similarity into verification-style acceptance and rejection rates.

4.5. Main Reporting Template

Table 2 uses the same grouped layout commonly adopted in purification papers: each protection method is expanded by cloning backend, the **Protected** columns report the no-purification control, and the remaining groups report post-purification xSVA/dSVA under four diffusion-based removal attacks. Numerical cells are intentionally left as run outputs rather than hand-written claims, so the table can be populated directly from experiment logs.

Protection Method	VC/TTS Method	Purification Method									
		Protected		AudioPure		DualPure		WavePurifier		De-AntiFake	
		xSVA	dSVA	xSVA	dSVA	xSVA	dSVA	xSVA	dSVA	xSVA	dSVA
AntiFake	CosyVoice	—	—	—	—	—	—	—	—	—	—
	F5-TTS	—	—	—	—	—	—	—	—	—	—
	Chatterbox	—	—	—	—	—	—	—	—	—	—
	Avg.	—	—	—	—	—	—	—	—	—	—
SafeSpeech	CosyVoice	—	—	—	—	—	—	—	—	—	—
	F5-TTS	—	—	—	—	—	—	—	—	—	—
	Chatterbox	—	—	—	—	—	—	—	—	—	—
	Avg.	—	—	—	—	—	—	—	—	—	—
E2E-VGuard	CosyVoice	—	—	—	—	—	—	—	—	—	—
	F5-TTS	—	—	—	—	—	—	—	—	—	—
	Chatterbox	—	—	—	—	—	—	—	—	—	—
	Avg.	—	—	—	—	—	—	—	—	—	—
AttackVC	CosyVoice	—	—	—	—	—	—	—	—	—	—
	F5-TTS	—	—	—	—	—	—	—	—	—	—
	Chatterbox	—	—	—	—	—	—	—	—	—	—
	Avg.	—	—	—	—	—	—	—	—	—	—
VoiceGuard	CosyVoice	—	—	—	—	—	—	—	—	—	—
	F5-TTS	—	—	—	—	—	—	—	—	—	—
	Chatterbox	—	—	—	—	—	—	—	—	—	—
	Avg.	—	—	—	—	—	—	—	—	—	—
PURGUARD	CosyVoice	—	—	—	—	—	—	—	—	—	—
	F5-TTS	—	—	—	—	—	—	—	—	—	—
	Chatterbox	—	—	—	—	—	—	—	—	—	—
	Avg.	—	—	—	—	—	—	—	—	—	—
Total	Avg.	—	—	—	—	—	—	—	—	—	—

TABLE 2. MAIN PURIFICATION-AWARE BENCHMARK IN THE GROUPED FORMAT USED BY PRIOR PURIFICATION WORK. EACH CELL REPORTS xSVA/dSVA AFTER CLONING FROM THE PROTECTED REFERENCE OR THE PURIFIED PROTECTED REFERENCE. LOWER VALUES INDICATE STRONGER RETAINED PROTECTION FROM THE DEFENDER’S PERSPECTIVE. RESULTS ARE TO BE POPULATED FROM EXPERIMENT LOGS FOR THE PROTECTION METHODS LISTED AT LEFT, INCLUDING PURGUARD.

4.6. Adaptive and Transfer Settings

To distinguish robustness against a fixed diffusion purifier from robustness against an adaptive attack chain, we include three transfer settings. The first changes the diffusion purifier while keeping the protected waveform fixed. The second changes the cloning backend after purification. The third changes the speaker verifier used to score the output. These tests measure whether PURGUARD overfits to the surrogate purifier or optimization encoders.

We also evaluate stochastic purification by varying random seeds and reporting either mean performance with dispersion or worst-case performance across seeds. This is necessary because a defense may look successful under one denoising sample while failing under another. Finally, we include the purified-clean control for every purifier. If a purifier damages clean speech identity, then poor cloning after purification cannot be credited to the protection method.

4.7. Ablations and Robustness Analyses

The core ablation compares three variants: **Speaker-only**, which removes $\mathcal{L}_{\text{traj}}$; **Trajectory-only**, which removes the post-purification identity term; and **PURGUARD**, which uses both terms. A second ablation varies α and β to

measure the trade-off between direct identity disruption and purification resistance. A third ablation varies the reverse timestep and capture stride to test whether robustness depends on a narrow diffusion schedule.

Because DiffBreak shows that stochastic and adaptive evaluation can overturn apparent robustness claims for diffusion purification, we report repeated purification samples or worst-case statistics over seeds [12]. We pair identity metrics with SNR, PESQ, and STOI so that improvements cannot be attributed merely to excessive audible distortion. Together, these experiments test the main claim of PURGUARD: once purification becomes part of the attack chain, robust voice protection should be optimized against the purifier itself.

4.8. Artifact and Reproducibility Plan

For reproducibility, we release perturbation-generation code, purifier wrappers, evaluation scripts, threshold-calibration code, configuration files, random seeds, and metadata for all generated protected samples. The artifact separates protected audio, purified audio, and cloned outputs so that each stage can be audited for identity loss. Because the work involves voice identity and impersonation risk, released audio follows dataset licenses and avoids examples that identify private speakers.

5. Conclusion

PURGUARD reframes voice anti-cloning as a purification-aware protection problem. Instead of assuming that a protected waveform is fed directly into a cloning system, we model the stronger and increasingly realistic attacker who first removes perturbations with a diffusion purifier. Under this threat model, protection must survive not only direct speaker extraction but also the attacker’s reverse denoising path.

PURGUARD addresses this gap by combining post-purification speaker disruption with trajectory-level deviation inside the purifier, and by providing a benchmark protocol with clean-purification controls and calibrated evaluator thresholds. Our analysis shows that trajectory supervision is a necessary complement to identity-only optimization: output-level gradients alone are insufficient when the purifier’s stochastic path can route around the perturbation.

Several directions remain open. First, the current framework uses a differentiable surrogate purifier during optimization and evaluates transfer to black-box target purifiers; extending beyond this to no-surrogate or stronger ensemble-purifier settings is an important practical step. Second, psychoacoustic masking and over-the-air robustness constraints should be incorporated to tighten the perceptual budget. Third, the arms race between purification and protection is ongoing: as bridge-diffusion purifiers such as VocalBridge become stronger, protection methods will need to track the evolving attack surface. We release our implementation and evaluation protocol to support reproducible progress on this problem.

6. Limitations

PURGUARD currently studies surrogate-based purification-aware optimization and evaluates whether the resulting protection transfers to black-box target purifiers. This captures the practical case where the defender can train against a local purifier but does not know the attacker’s deployed purifier; it does not fully cover no-surrogate defenders, learned purifier ensembles, or attackers who jointly adapt purification and synthesis. The current implementation also focuses on waveform-domain perturbations with bounded amplitude; future work should add psychoacoustic masking, over-the-air transformations, and human listening studies. We plan to extend the benchmark with broader stochastic purification trials before drawing claims across all purifiers and cloning backends.

7. Ethical Considerations

This work studies defenses against unauthorized voice cloning and therefore involves dual-use capabilities. The intended use of PURGUARD is to help speakers publish audio with reduced risk of impersonation after purification-based removal attacks. The evaluation should use public or consented speech datasets, report aggregate metrics, and

avoid releasing cloned examples that could be misused for impersonation. We do not claim that perturbation-based protection replaces policy, watermarking, authentication, or platform-side abuse detection; rather, it provides a data-source defense that can be combined with these safeguards.

8. LLM Usage Statement

Large language models were used as writing assistants for organization, editing, and wording suggestions. All technical claims, citations, experiments, and final text remain the responsibility of the authors.

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